

Movie Rental Analytics

MECE Breakdown · Areas · Sub-Areas · KPIs · Data Sources

Main Dashboard	Customers	Payments
Rentals	Inventory	

This document provides a structured MECE breakdown of the Sakila DVD Rental analytics framework, mapping each business domain to measurable KPIs and the underlying database tables required for Power BI report development.

MAIN DASHBOARD

Sub-Area	KPIs / Metrics	Data Sources
Revenue Summary	Total Revenue, Store 1 Revenue, Store 2 Revenue, Avg Payment	payment
Rental Overview	Total Rentals, Not Returned Count, Avg Rental Duration (days)	rental + inventory
Customer Snapshot	Total Customers, Active Customers, Inactive Customers, Avg Spend / Customer	customer + payment
Revenue Trend	Monthly Revenue Line Chart — performance trajectory over selected period	payment (payment_date hierarchy)
Staff Performance	Revenue by Staff — bar chart comparing staff member contributions	payment + staff
Top Films	Top 7 Rented Films — ranked by rental frequency	rental + film
Top Customers	Top 8 Customers by Revenue — ranked by total spend	payment + customer
Rentals by Genre	Rentals per Category — distribution of rentals across film genres	rental + film + category
Interactive Filters	Slicers for Date (Month), Store, Staff — dynamic cross-filtering	payment + rental + staff

CUSTOMERS

Sub-Area	KPIs / Metrics	Data Sources
Customer Volume	Total Customers, Active Customers, Inactive Customers	customer
Spend Analysis	Avg Spend / Customer, Total Revenue, MAX Single Payment	payment + customer
Customer Segmentation	Total Spend by Customer Segment — grouped bar/donut by segment label	customer (Customer Segment) + payment
Top Customers	Top 8 Customers by Revenue — ranked leaderboard with spend values	payment + customer (first_name)
Geographic Distribution	Customer Count & Revenue by City — regional spread of customer base	customer + city
Payment Buckets	Payment distribution by Payment Bucket — segmented spend tiers	payment (Payment Bucket field)
Revenue Trend	Monthly Revenue Trend by Customer — time-series line chart	payment (payment_date hierarchy)
Film Rating Preference	Rentals / Revenue by Film Rating — audience type breakdown (G, PG, R, etc.)	rental + film (rating)

PAYMENTS

Sub-Area	KPIs / Metrics	Data Sources
Revenue Performance	Total Revenue, TOTAL PAYMENTS count, Avg Payment, MAX Single Payment	payment
Store Revenue Split	Store 1 Revenue vs Store 2 Revenue — side-by-side comparison	payment + inventory (store_id)
Monthly Revenue Trend	Revenue Trend (Monthly) — line/bar chart over payment_date hierarchy	payment (payment_date.Month)
Payment Segmentation	Payments by Payment Bucket — distribution of transaction size bands	payment (Payment Bucket)
Staff Revenue	Revenue by Staff — individual staff member revenue attribution	payment + staff (first_name)
Customer Revenue Ranking	Top 8 Customers by Revenue — highest-value customer identification	payment + customer (first_name)
Category Revenue Mix	Revenue / Rentals by Film Category — genre-level revenue contribution	payment + rental + film + category

RENTALS

Sub-Area	KPIs / Metrics	Data Sources
Rental Volume	Total Rentals, Not Returned Count, Avg Rental Duration (days)	rental
Rental Trend	Monthly Rental Volume — time-series count over rental_date hierarchy	rental (rental_date.Month)
Genre Demand	Rentals by Genre (Category) — top genres by rental frequency	rental + film + category
Film Popularity	Top 7 Rented Films — most frequently rented titles	rental + film (title)
Return Status	Not Returned Items — count of active/unreturned rentals, overdue risk	rental (return_date null check)
Rental Duration Analysis	Avg Rental Duration (days) by film/category — operational hold time	rental + film
Store Rental Distribution	Rentals per Store — volume comparison between Store 1 and Store 2	rental + inventory (store_id)
Customer Rental Frequency	Rentals per Customer — repeat rental behaviour and loyalty signals	rental + customer

INVENTORY

Sub-Area	KPIs / Metrics	Data Sources
Inventory Status	Total Inventory Items, Unique Films Available, Currently Rented Out, Available Right Now	inventory
Inventory by Category	Stock count per film category — horizontal bar chart showing category depth	inventory + film + category (name)
Inventory by Film Rating	Stock distribution by Film Rating (G/PG/PG-13/R/NC-17) — donut chart	inventory + film (rating)
Store Stock Balance	Inventory per Store — clustered bar comparing stock between Store 1 & Store 2	inventory (store_id)
Top Stocked Films	Top 10 Most Stocked Films — identifies potential overstock titles	inventory + film (title)
Rental Duration Stock	Inventory by Rental Duration — stock count by film rental_duration setting	inventory + film (rental_duration)
Inventory Utilisation	Inventory Utilisation Rate — % of stock currently rented vs available	inventory + rental